

DW_bc_1

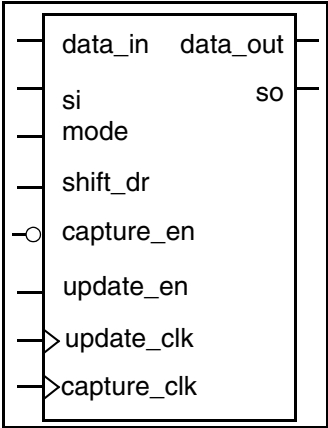
Boundary Scan Cell Type BC_1

Version, STAR and Download Information: [IP Directory](#)

Features and Benefits

- IEEE Standard 1149.1 compliant
- Synchronous or asynchronous scan cells with respect to t_{ck}
- Supports the standard instructions: EXTEST, SAMPLE/PRELOAD, and BYPASS
- Supports the optional instructions INTEST, RUNBIST, CLAMP, and HIGHZ

Revision History



Description

DW_bc_1 is a boundary scan cell that can be used as a system input cell or as an output cell. When used as an input cell, DW_bc_1 provides control over the inputs to the core logic of the IC. When used as an output cell, DW_bc_1 provides control over the IC output signals. The Boundary Scan Description Language (BSDL) description of this cell is of type BC_1 described in the BSDL package STD_1149_1_1990.

The DW_bc_1 cell can be synchronous or asynchronous with respect to t_{ck} (Test Clock system pin), depending on the port connections.

The `mode` signal gives the Test Access Port (TAP) instructions control of the boundary scan cell.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
capture_clk	1 bit	Input	Clocks data into the capture stage
update_clk	1 bit	Input	Clocks data into the update stage
capture_en	1 bit	Input	Enable for data clocked into the capture stage, active low
update_en	1 bit	Input	Enable for data clocked into the update stage, active high
shift_dr	1 bit	Input	Enables the boundary scan chain to shift data one stage toward its serial output (tdo)
mode	1 bit	Input	Determines whether <code>data_out</code> is controlled by the boundary scan cell or by the <code>data_in</code> signal
si	1 bit	Input	Serial path from the previous boundary scan cell
data_in	1 bit	Input	Input data

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
data_out	1 bit	Output	Output data
so	1 bit	Output	Serial path to the next boundary scan cell

Table 1-2 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare or Test-IEEE-STD-1149-1

Table 1-3 Simulation Models

Model	Function
DW04.DW_BC_1_CFG_SIM	Design unit name for VHDL simulation
dw/dw04/src/DW_bc_1_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_bc_1.v	Verilog simulation model source code

Table 1-4 lists the required values of the `mode` signal for each of the TAP instructions that DW_bc_1 supports.

Table 1-4 Mode Signal Generation for DW_bc_1

Instruction	Mode for Input Cell	Mode for Output Cell
EXTEST	1	1
SAMPLE/PRELOAD	0	0
INTEST	1	1 ^a
CLAMP	X	1
RUNBIST	X	1 ^a
BYPASS	0	0

- a. If you do not want these instructions to drive the output pins with pre-loaded data held in the boundary scan register, then these instructions are not needed to determine the state of the `mode` signal. Instead, the instruction must be added to the output enable logic to force every system output pin to an inactive drive state.

Table 1-5 lists the connections for asynchronous boundary scan chains.

Table 1-5 Port Connections for Asynchronous Boundary Scan Chains

DW_bc_1 Port Name	Connection
capture_clk	clock_dr from TAP controller
update_clk	update_dr from TAP controller
capture_en	Logic zero
update_en	Logic one
shift_dr	shift_dr from TAP controller
mode	Mode generation logic
si	so from previous boundary scan cell
data_in	System input pin for input cells or IC output logic for output cells
data_out	IC input logic for input cells or system output pin for output cells
so	si of next boundary scan cell

Table 1-6 lists the connections for synchronous boundary scan chains.

Table 1-6 Port Connections for Synchronous Boundary Scan Chains

DW_bc_1 Port Name	Connection
capture_clk	tck from system pin
update_clk	tck_n from system pin
capture_en	sync_capture_en from TAP controller
update_en	sync_update_dr from TAP controller
shift_dr	shift_dr from TAP controller
mode	Mode generation logic
si	so from previous boundary scan cell
data_in	System input pin for input cells or IC output logic for output cells
data_out	IC input logic for input cells or system output pin for output cells
so	si of next boundary scan cell

Related Topics

- [Application Specific - JTAG Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW_bc_1_inst is
  port (inst_capture_clk : in std_logic;  inst_update_clk  : in std_logic;
        inst_capture_en  : in std_logic;  inst_update_en   : in std_logic;
        inst_shift_dr    : in std_logic;  inst_mode        : in std_logic;
        inst_si          : in std_logic;  inst_data_in     : in std_logic;
        data_out_inst    : out std_logic; so_inst          : out std_logic );
end DW_bc_1_inst;

architecture inst of DW_bc_1_inst is
begin
  -- Instance of DW_bc_1
  U1 : DW_bc_1
    port map (capture_clk => inst_capture_clk,
              update_clk => inst_update_clk,  capture_en => inst_capture_en,
              update_en => inst_update_en,    shift_dr => inst_shift_dr,
              mode => inst_mode,    si => inst_si,    data_in => inst_data_in,
              data_out => data_out_inst,    so => so_inst );
end inst;

-- pragma translate_off
configuration DW_bc_1_inst_cfg_inst of DW_bc_1_inst is
  for inst
  end for; -- inst
end DW_bc_1_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_bc_1_inst(inst_capture_clk, inst_update_clk, inst_capture_en,
                    inst_update_en, inst_shift_dr, inst_mode, inst_si,
                    inst_data_in, data_out_inst, so_inst );

    input inst_capture_clk;
    input inst_update_clk;
    input inst_capture_en;
    input inst_update_en;
    input inst_shift_dr;
    input inst_mode;
    input inst_si;
    input inst_data_in;
    output data_out_inst;
    output so_inst;

    // Instance of DW_bc_1
DW_bc_1
    U1 (.capture_clk(inst_capture_clk), .update_clk(inst_update_clk),
        .capture_en(inst_capture_en), .update_en(inst_update_en),
        .shift_dr(inst_shift_dr), .mode(inst_mode), .si(inst_si),
        .data_in(inst_data_in), .data_out(data_out_inst), .so(so_inst) );
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 4 ■ Added this Revision History table and the document links on this page

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